

PRAKTIKUM SISTEM OPERASI

MODUL 5

EKSPLORASI PROSES



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

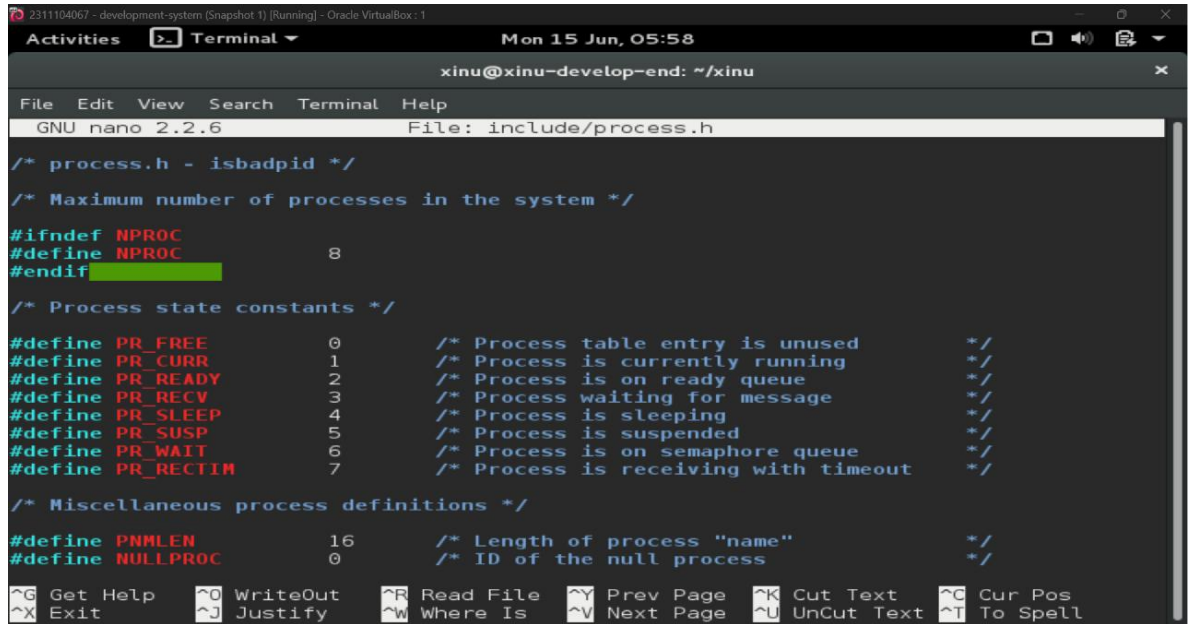
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL/TP

1. [10 Poin] Jawablah pertanyaan berikut ini:



```
xinu@xinu-develop-end: ~/xinu
GNU nano 2.2.6 File: include/process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */
```

- Berapa banyaknya maksimum proses yang ada pada Xinu?
8
- Berapa maksimal panjang nama suatu proses pada Xinu?
16
- Berapa nilai prioritas awal pada saat proses dibuat?

```
include/process.h:#define INITPRIO 20 /* Initial process priority */
```

- Berapa total state pada Xinu? Sebutkan! 8 state

```
/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */
```

```
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ ls
compile  COPYRIGHT  device  lib      mutex.c  shell      system
config  DESCRIPTION include  main.c  net      signaling.c xsh_mycmd.c
xinu@xinu-develop-end:~/xinu$ nano include/process.h
xinu@xinu-develop-end:~/xinu$ nano system/create.c
xinu@xinu-develop-end:~/xinu$ grep -R "INITPRI0"
include/process.h:#define      INITPRI0      20      /* Initial process priority */
/
system/initialize.c:      resume(create((void *)startup, INITSTK, INITPRI0,
system/initialize.c:      resume(create((void *)main, INITSTK, INITPRI0,
xinu@xinu-develop-end:~/xinu$ nano include/process.h
xinu@xinu-develop-end:~/xinu$
```

2. [20 Poin] Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!
 - a. Menjalankan find . -name "xsh_ps.c" di terminal

```
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_ps.c"
./shell/xsh_ps.c
```

- b. Setelah muncul alamat filenya, buka sesuai alamat file yaitu
./shell/xsh_ps.c

```
GNU nano 2.2.6                               File: sheli
/* xsh_ps.c - xsh_ps */

#include <linux.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_ps - shell command to print the process table
 *-----
 */
shelliced xsh_ps(int nargs, char *args[])
{
    struct proctent *prptr;          /* Pointer to process          */
    int32 i;                        /* Index into proctabl       */
    char *prstate[] = {             /* Names for process states  */
        "free ", "curr ", "ready", "recv ", "sleep", "susp ",
        "wait ", "rtime"};

    /* For argument '--help', emit help about the 'ps' command */
    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("User: %s\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays information about running processes\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }

    /* Check for valid number of arguments */
    if (nargs > 1) {
        fprintf(stderr, "%s: too many arguments\n", args[0]);
        fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
        return 1;
    }

    /* Print header for items from the process table */
    printf("%3s %16s %3s %4s %10s %10s %10s\n",
        "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
        "Stack Ptr", "Stack Size");

    printf("%3s %16s %3s %4s %10s %10s %10s\n",
        "----", "-----", "-----", "----", "-----", "-----",
        "-----", "-----");

    /* Output information for each process */
    for (i = 0; i < NPROC; i++) {
        prptr = &proctab[i];
        if (prptr->prstate == PR_FREE) { /* Skip unused slots */
            continue;
        }
        printf("%3d %16s %s %4d %10d 0x%08X 0x%08X %8d\n",
            i, prptr->prname, prstate[(int)prptr->prstate],
            prptr->prprio, prptr->pparent, prptr->prstackbase,
            prptr->prstackptr, prptr->prstacklen);
    }
    return 0;
}
```

- c. Menambahkan pada 20 baris terakhir di source code:

```

        printf("\tDisplays information about running processes\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }

    /* Check for valid number of arguments */
    if (nargs > 1) {
        fprintf(stderr, "%s: too many arguments\n", args[0]);
        fprintf(stderr, "Try '%s --help' for more information\n",
                args[0]);
        return 1;
    }
    /* Menampilkan stack proses */
    /* Print header for items from the process table */

    printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
           "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
           "Stack Ptr", "Stack Size");

    printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
           "-----", "-----", "-----", "-----", "-----", "-----",
           "-----", "-----");

    /* Output information for each process */
    for (i = 0; i < NPROC; i++) {
        prptr = &proctab[i];
        if (prptr->prstate == PR_FREE) { /* Skip unused slots */
            continue;
        }
        printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
               i, prptr->prname, prstate[(int)prptr->prstate],
               prptr->prprio, prptr->prparent, prptr->prstkbase,
               prptr->prstkptr, prptr->prstklen);
    }
    return 0;
}

```

3. [35 Poin] Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

```
xsh $ ps
```

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size	Msg	Content
0	pmull	ready	0	0	0x005FDFFC	0x00FFFF04	8192	1	4
1	rdsproc	susp	200	0	0x005FBFFC	0x005F8FC8	16384	0	0
2	ipout	wait	500	0	0x005F7FFC	0x005F7F40	8192	0	0
3	netin	wait	500	0	0x005F5FFC	0x005F5EE0	8192	0	0
5	Main process	recv	20	4	0x005E3FFC	0x005E3F64	65536	0	0
6	shell	recv	50	5	0x005D3FFC	0x005D3C7C	8192	0	0
7	ps	curr	20	6	0x005F3FFC	0x005F3DC8	8192	0	0

Kolom Msg adalah banyaknya pesan yang ada dalam proses.

Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c

a. source code asli:

```
/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "----", "----", "-----",
       "-----", "-----");
```

b. Source code setelah dimodifikasi:

```
printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %8s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "----", "----", "-----",
       "-----", "-----", "-----", "-----");
```

- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya

```
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$ make
```

- Jalankan Backend VM

```
iPXE (PCI E2:00.0) starting execution...ok
iPXE initialising devices...ok

iPXE 1.21.1 -- Open Source Network Boot Firmware -- http://ipxe.org
Features: DNS TFTP PXE PXEXT

net0: 08:00:27:c6:50:8a using pcnet32 on 0000:00:03.0 (open)
[Link:up, TX:0 TXE:0 RX:0 RXE:0]
Configuring (net0 08:00:27:c6:50:8a)..... ok
net0: 192.168.201.115/255.255.255.0
Next server: 192.168.201.1
Filename: xinu.grub
tftp://192.168.201.1/xinu.grub... ok
xinu.grub : 131861 bytes [PXE-NBP]
Welcome to GRUB!
```

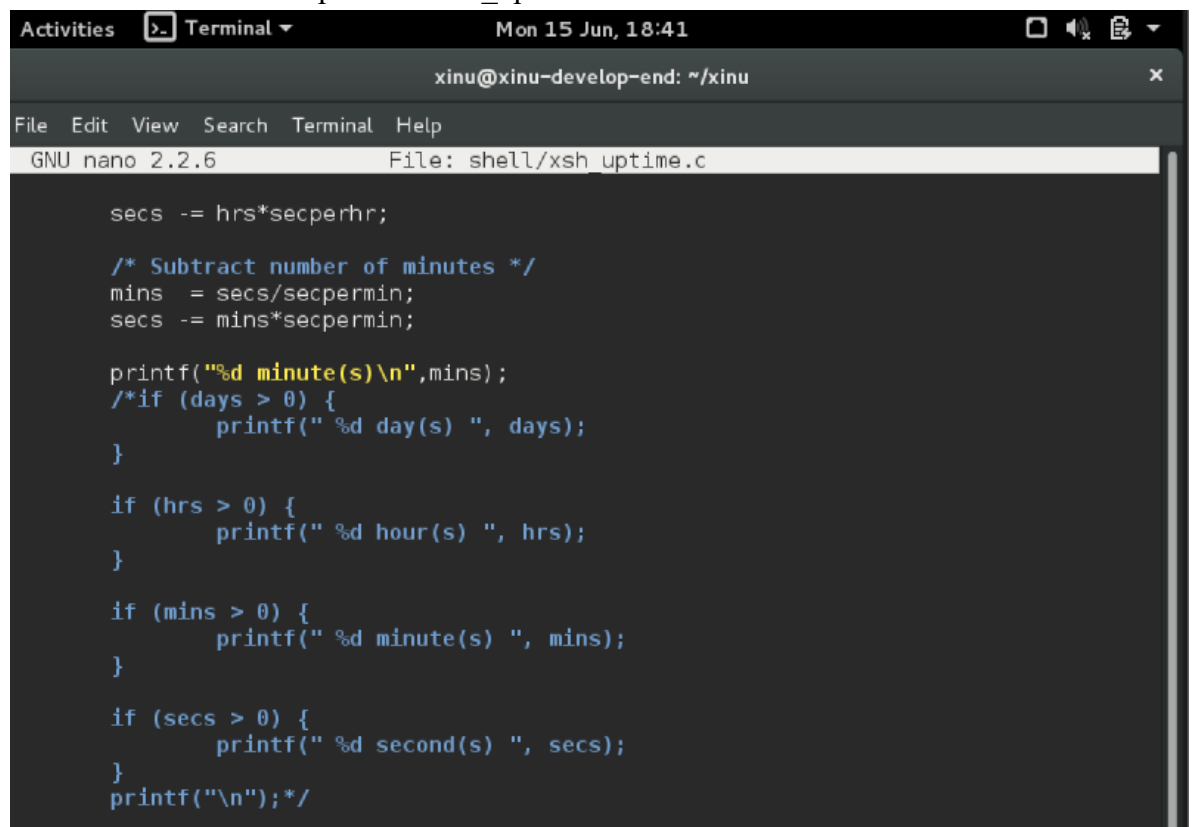
- Setelah sistem berjalan, jalankan perintah \$ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.
- Screenshot source kode dan output akhir hasil modifikasi=

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv   50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps              curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0
xsh $
```

4. [35 Poin] Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c



```

Activities  Terminal  Mon 15 Jun, 18:41
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: shell/xsh_uptime.c

secs -= hrs*secperhr;

/* Subtract number of minutes */
mins = secs/secpermin;
secs -= mins*secpermin;

printf("%d minute(s)\n",mins);
/*if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");*/

```

- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ nano shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$
xinu@xinu-develop-end:~/xinu$ make clean
make: *** No rule to make target 'clean'. Stop.
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$ make
making the config program
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
flex config.l
bison -y          config.y
gcc -o config y.tab.c -lfl
./config Configuration conf.c conf.h
cp -p conf.h ../include
cp -p conf.c ../system
make[1]: Leaving directory '/home/xinu/xinu/config'
making ../system/conf.c
make[1]: Entering directory '/home/xinu/xinu/compile'
forcing a rebuild of conf.h and conf.c
make[2]: Entering directory '/home/xinu/xinu/compile'

```

- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan

- Screenshot source code dan output akhir hasil modifikasi

```

Activities  Terminal  Mon 15 Jun, 18:37
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

115948 bytes of Xinu code.
[0x00100000 to 0x0011C4EB]
148424 bytes of data.
[0x0011FF00 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a09030f)

-----Berlian Seva Astryana-----

XINU

-----2311104067-----

Welcome to Xinu!

sh $ uptime
0 minute(s)
sh $ xinu
command xinu not found
sh $ ps

```

Pid	Name	State	Prio	Msg	Content	Ppid	Stack	Base	Stack	Size
0	prnull	ready	0	1	0x00000000 0x00000000	6283260				
1	ipout	wait	500	0	0x00000000 0x00000000	6275068				
2	netin	wait	500	0	0x00000000 0x00000000	6266876				
4	Main process	recv	20	0	0x00000000 0x00000003	6193148				
5	shell	recv	50	0	0x00000006 0x00000004	6127612				
7	ps	curr	20	0	0x00000000 0x00000005	6258684				

```

sh $
TRI-4.7 for help | 115209 RNI | MCR | Minicos 2.7 | VT102 | Online 0:2 | ttu50

```